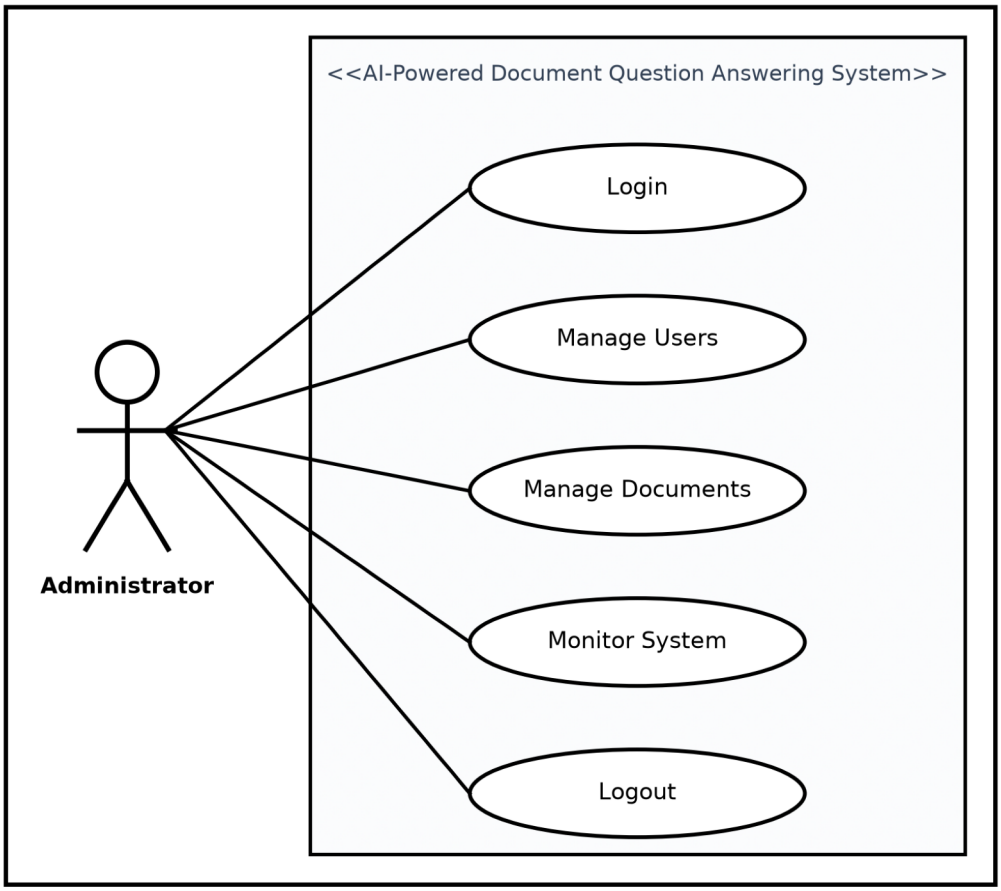
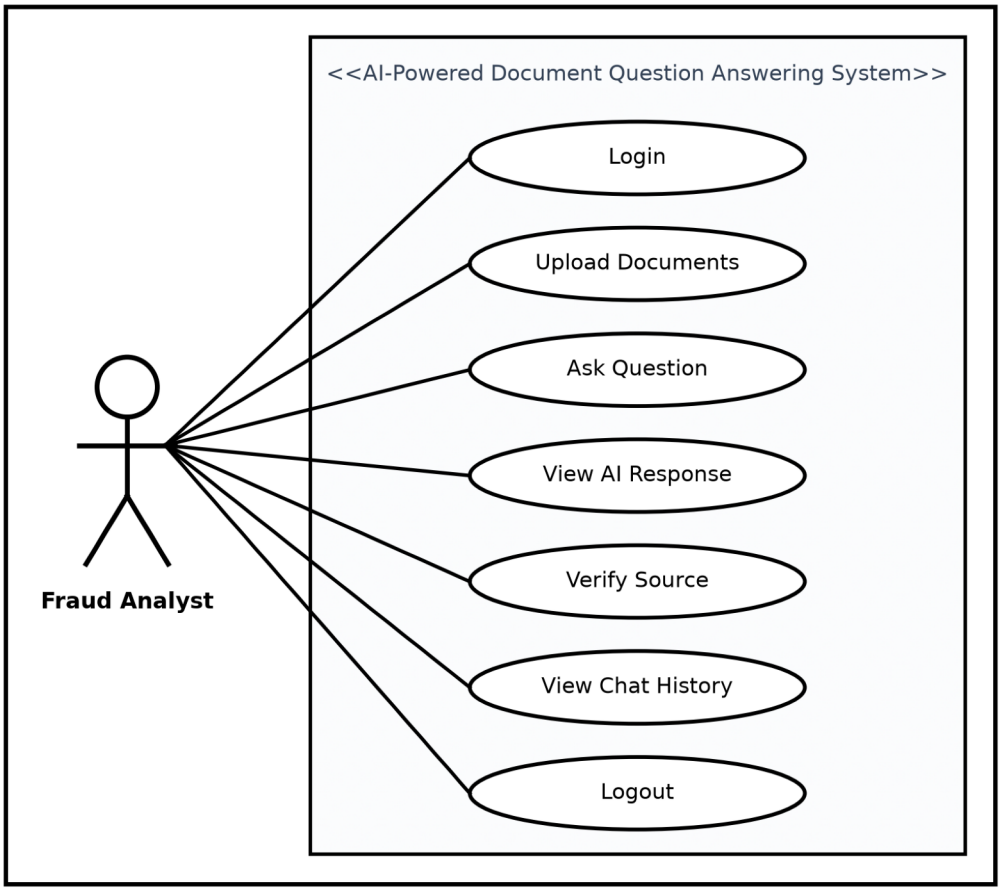
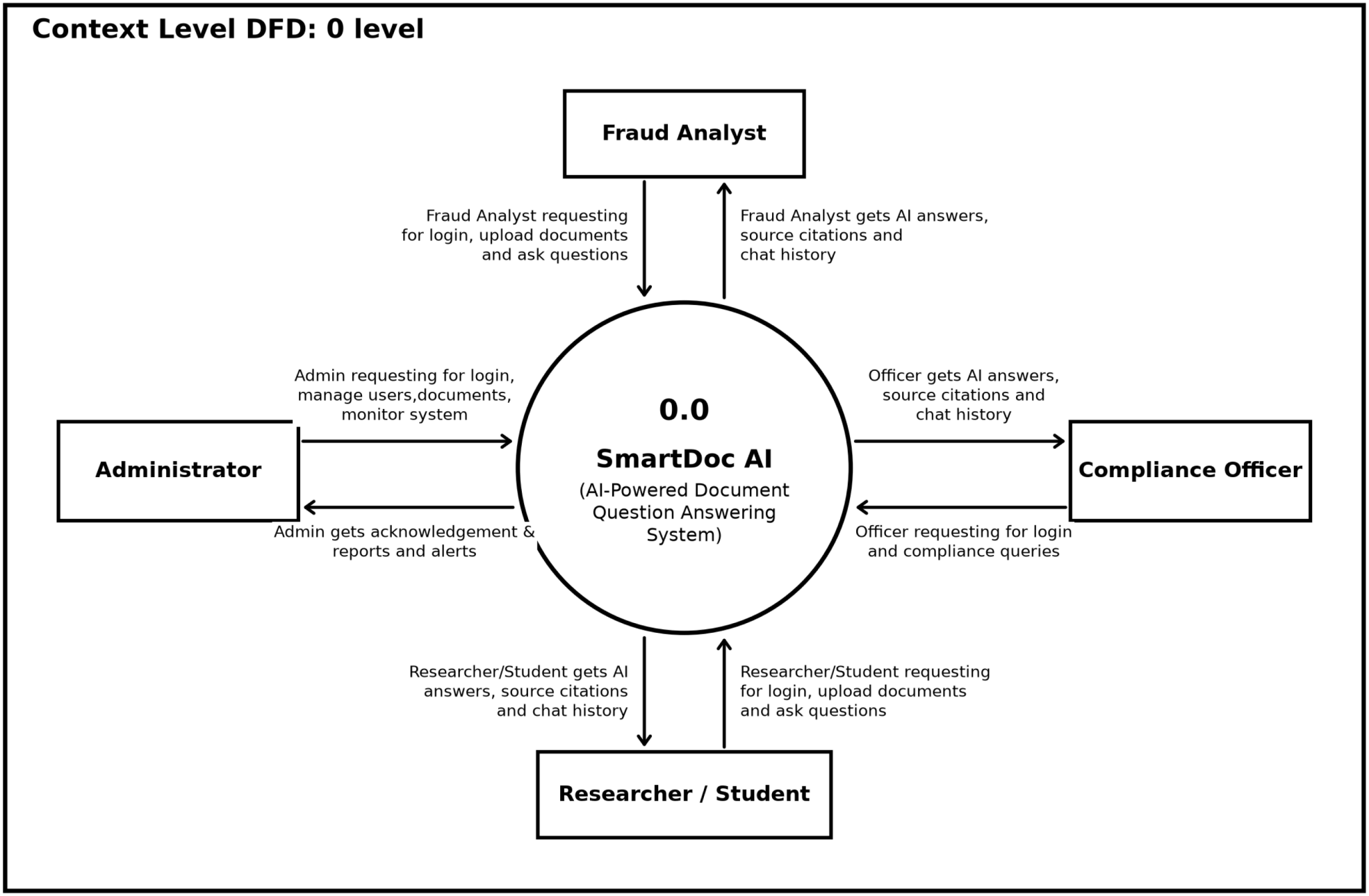


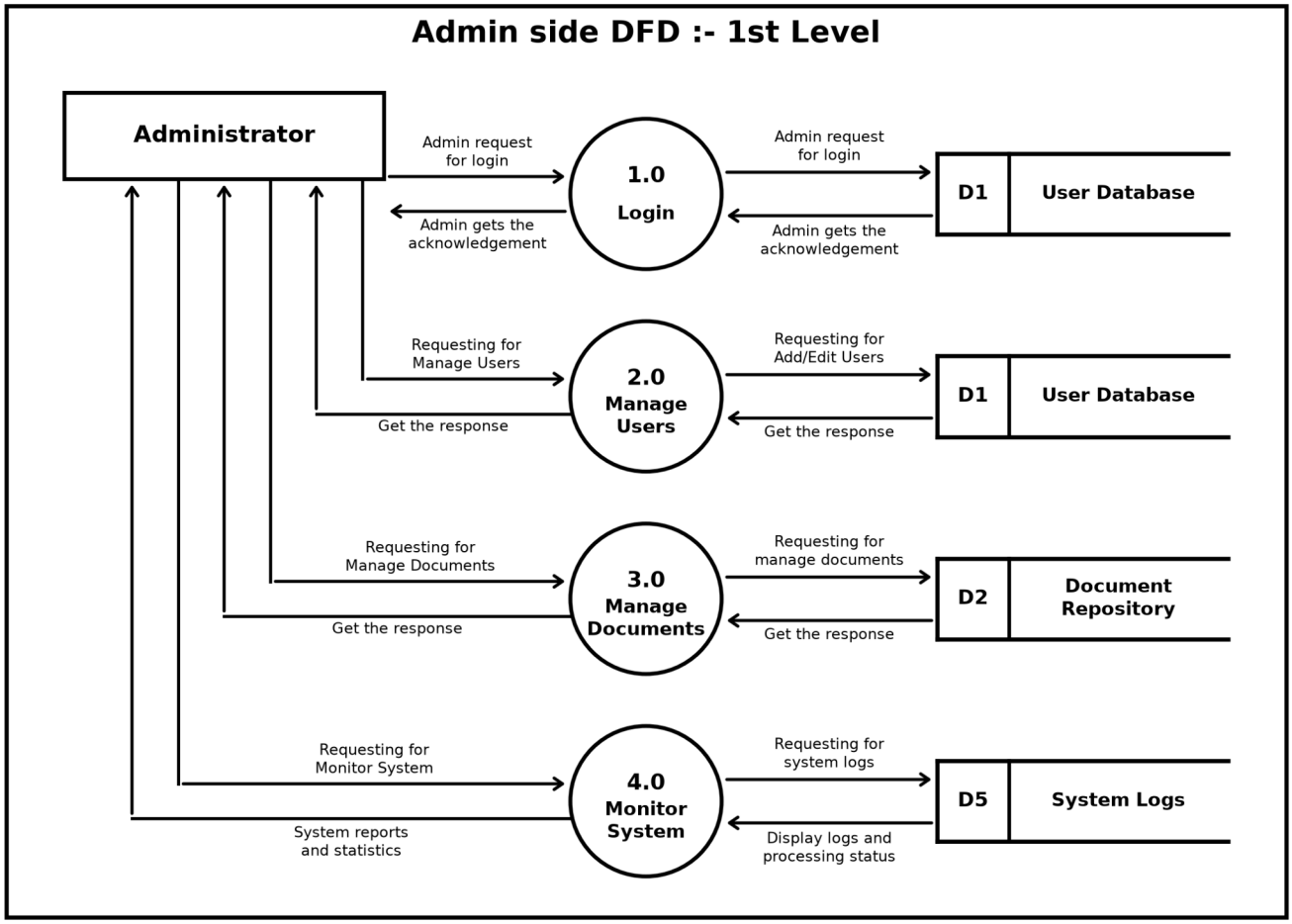
Use Case Diagram - Administrator



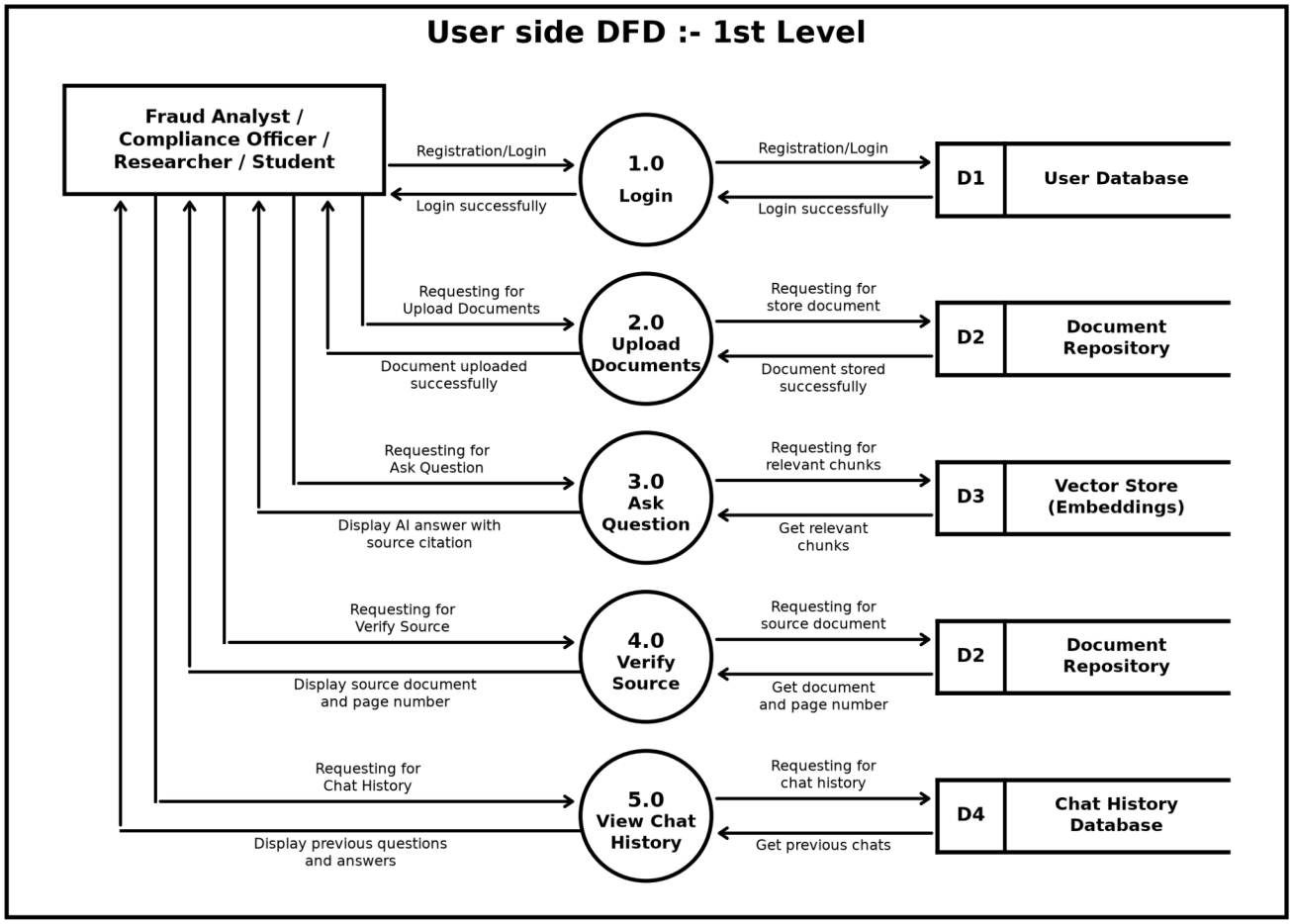
Use Case Diagram - Fraud Analyst



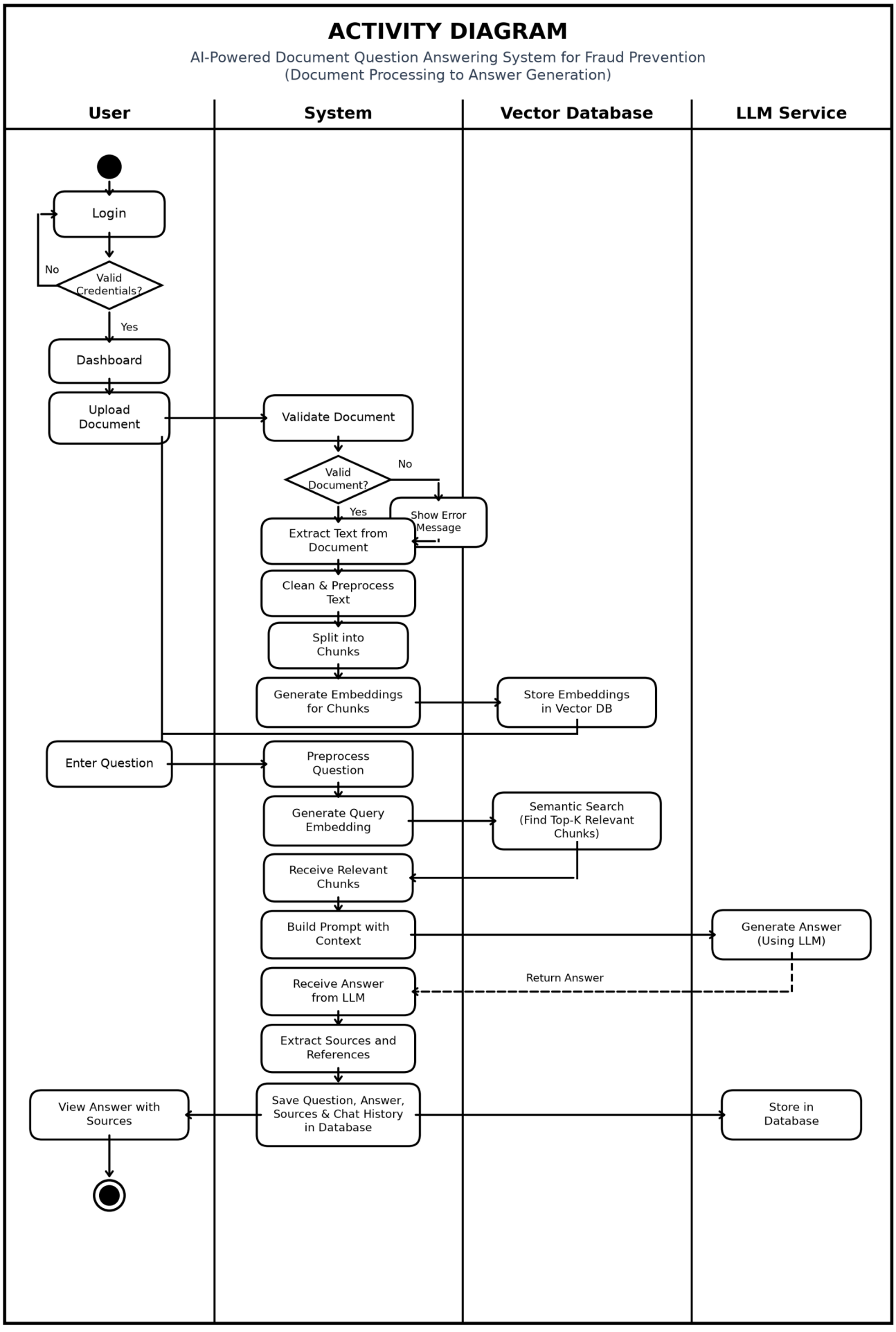
Context Level DFD: 0 Level

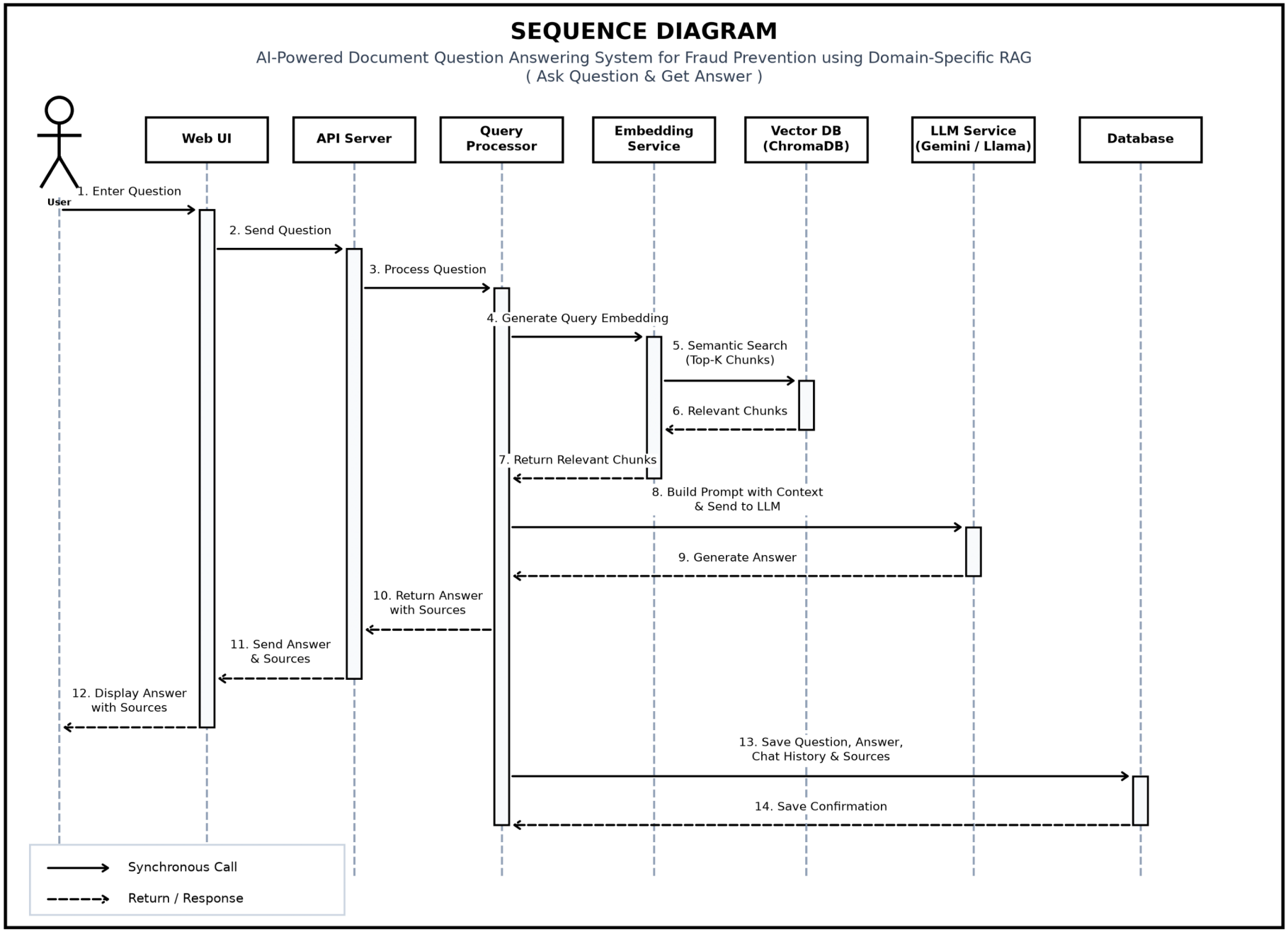


Admin Side DFD: 1st Level



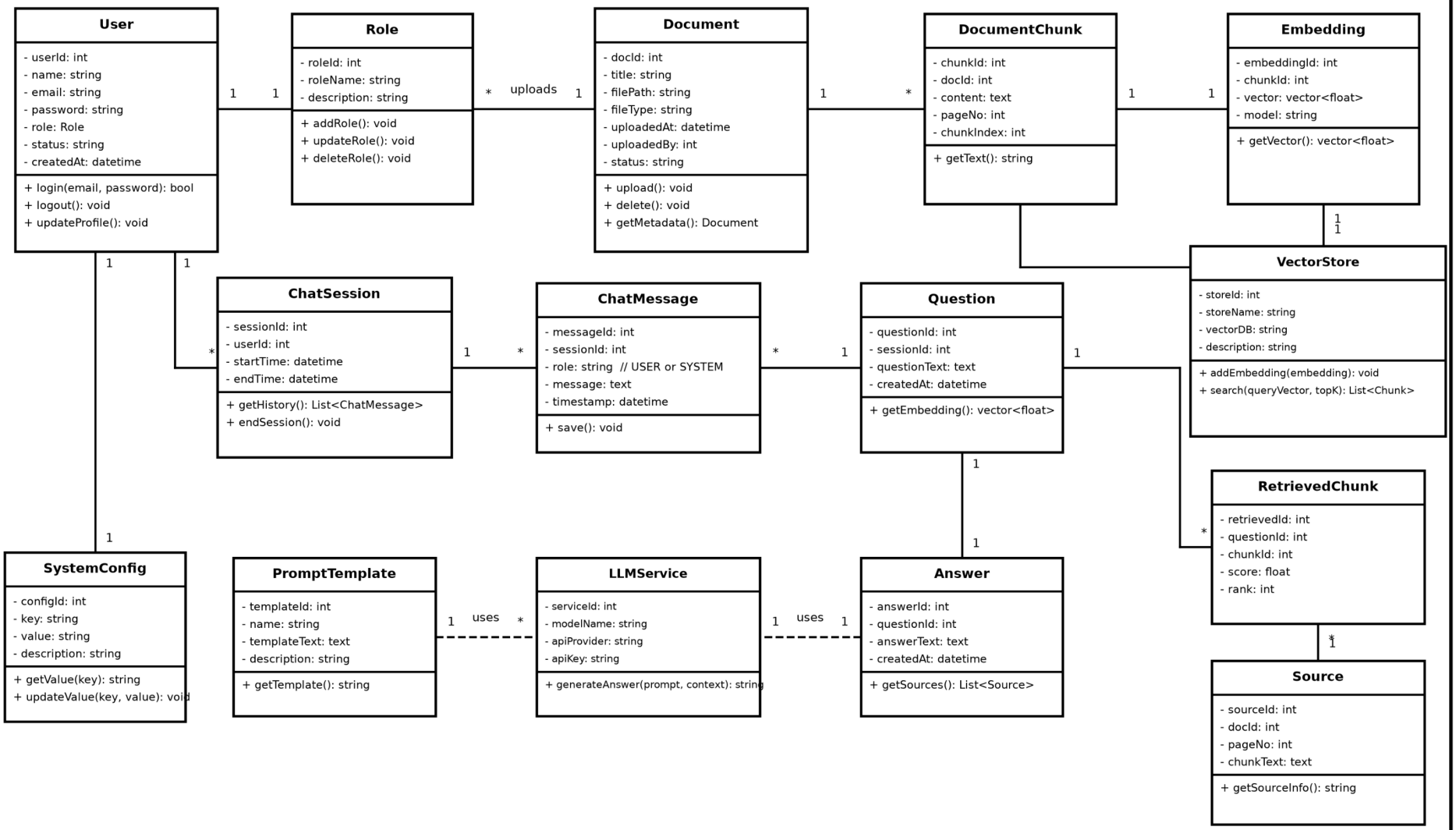
User Side DFD: 1st Level


ACTIVITY DIAGRAM
AI-Powered Document Question Answering System for Fraud Prevention
(Document Processing to Answer Generation)
User
System
Vector Database
LLM Service
Login
No
Valid Credentials?
Yes
Dashboard
Upload Document
Validate Document
Valid Document?
No
Yes
Show Error Message
Extract Text from Document
Clean & Preprocess Text
Split into Chunks
Generate Embeddings for Chunks
Store Embeddings in Vector DB
Enter Question
Preprocess Question
Generate Query Embedding
Semantic Search (Find Top-K Relevant Chunks)
Receive Relevant Chunks
Build Prompt with Context
Generate Answer (Using LLM)
Return Answer
Receive Answer from LLM
Extract Sources and References
View Answer with Sources
Save Question, Answer, Sources & Chat History in Database
Store in Database


SEQUENCE DIAGRAM
AI-Powered Document Question Answering System for Fraud Prevention using Domain-Specific RAG
(Ask Question & Get Answer)
User
Web UI
API Server
Query Processor
Embedding Service
Vector DB (ChromaDB)
LLM Service (Gemini / Llama)
Database
1. Enter Question
2. Send Question
3. Process Question
4. Generate Query Embedding
5. Semantic Search (Top-K Chunks)
6. Relevant Chunks
7. Return Relevant Chunks
8. Build Prompt with Context & Send to LLM
9. Generate Answer
10. Return Answer with Sources
11. Send Answer & Sources
12. Display Answer with Sources
13. Save Question, Answer, Chat History & Sources
14. Save Confirmation
Synchronous Call
Return / Response

CLASS DIAGRAM

AI-Powered Document Question Answering System for Fraud Prevention
using Domain-Specific RAG



4.2.2 Entity Relationship Diagram (ERD)

The relational schema of ARES is centered around the primary assessments entity, which maintains one-to-many relationships with child entities capturing findings, activity logs, generated reports, attack surfaces, and pentest actions.

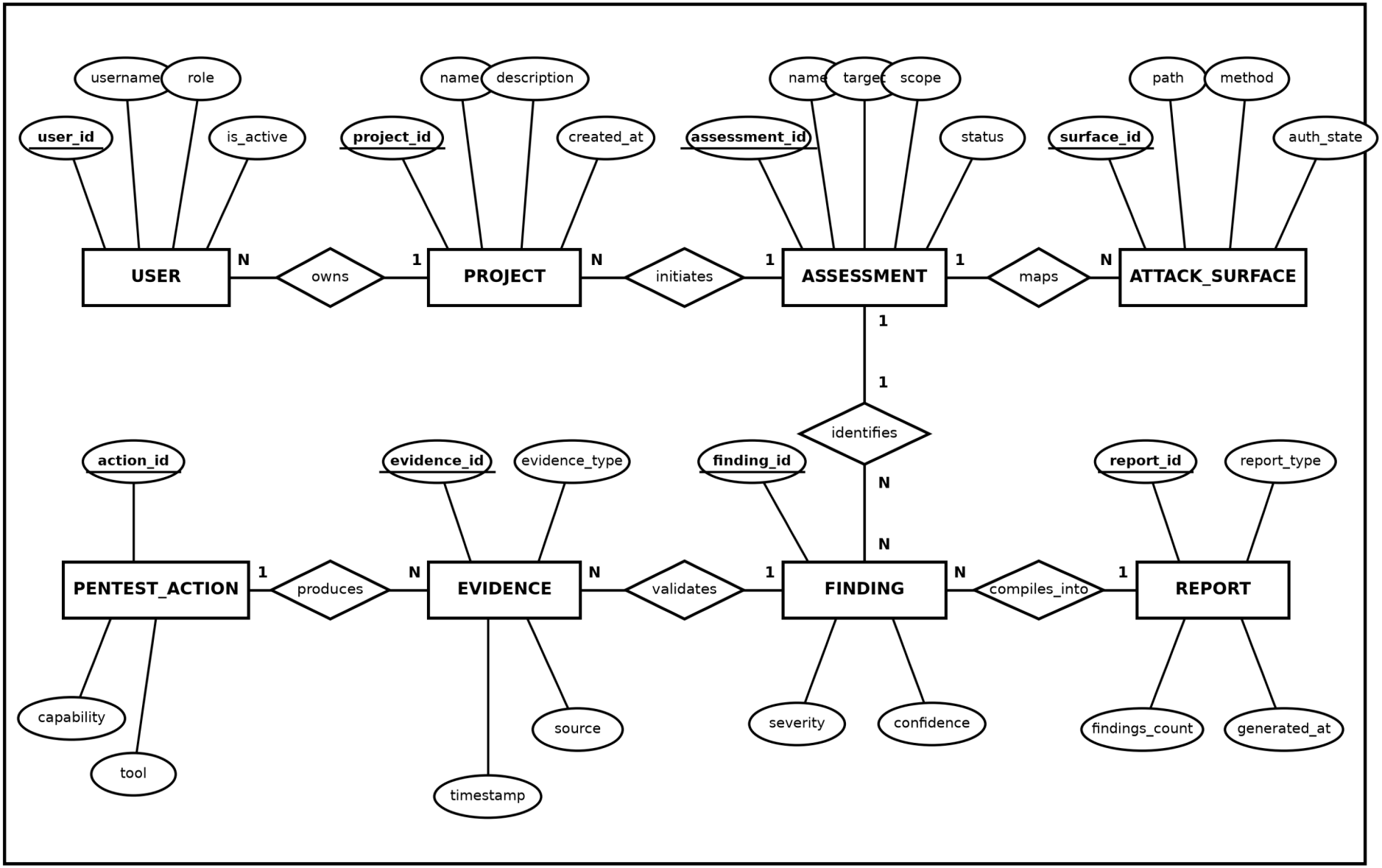


Fig 4.8: Entity Relationship Diagram (ERD) of ARES Database Schema